Kishan Edimadakala

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SUMMARY

Data Engineer with 4+ years of experience designing scalable data platforms, governed semantic layers, and applied AI/agentic systems. Proven track record in orchestrating metadata-driven ingestion, self-healing data observability, and reliable LLM-powered workflows (RAG, LangGraph, Bedrock) grounded in deterministic data sources. Skilled across distributed architectures (Databricks, Delta Lake, Snowflake, AWS, Azure/Fabric concepts), data contracts, and cross-functional leadership across product, applied science, and analytics teams.

EXPERIENCE

Data Engineer, *Amazon (Amazon Photos)* 12/2024 – Present

- Built and owned scalable data pipelines and warehouse architectures using AWS and Snowflake (Python, PySpark, SQL, Airflow) for a product serving 100M+ users, delivering governed, analytics-ready datasets.
- Architected scalable, metadata-driven data platforms and event-driven ingestion frameworks (AWS Glue, Lambda, Step Functions, EventBridge) processing 10 to 500+ TB of telemetry and multi-modal data.
- Built and deployed autonomous agentic and LLM tooling using AWS Bedrock, LangChain, and LangGraph to enable natural language querying across internal datasets, implementing evaluation guardrails to ensure reports remain grounded in deterministic source data.
- Designed dimensional models, data contracts, and semantic layers across Redshift, Snowflake, and Delta/Parquet, reducing ad hoc query costs by 30% via serverless architectures while preserving strict SLA performance for compliance and business intelligence.
- Implemented end-to-end self-healing observability, SLA monitoring, and automated incident recovery using DLQs, CloudWatch alerting, and Terraform/CDK infrastructure-as-code.
- Built the Photos Curated Memories system to run engagement scoring and scenario analysis, partnering closely with Product, Data Science, and Growth teams to turn ambiguous requirements into durable platform capabilities.
- Delivered the automated GDPR Profile Deletion compliance pipeline, establishing strict data governance and audit-ready data validation with zero production findings.
- Built GenAI/LLM-powered data tools (LangChain, LangGraph, RAG, FAISS, Pinecone) and mentored team members on GenAI integration practices.

Data Engineer, *Raven Software Solutions* 01/2024 – 11/2024

- Built and optimized ETL/ELT pipelines and data models (Databricks, Airflow, dbt, Kafka) on Azure and GCP, processing structured and unstructured data into Delta Lake-backed, governed, analytics-ready datasets.
- Modernized and unified data platforms by migrating legacy SQL pipelines to Databricks Lakehouse (Delta Lake, PySpark, Unity Catalog), creating a governed, single source of truth across enterprise datasets.
- Designed dynamic, parameterized data orchestration components that eliminated bespoke ETL engineering and accelerated new dataset onboarding by 40%.
- Enforced platform reliability and data quality guardrails using dbt tests, Monte Carlo, and Unity Catalog lineage, establishing automated remediation and anomaly detection.
- Led architecture design reviews and mentored junior engineers on dimensional modeling, CI/CD pipeline automation, and data contract enforcement.
- Built SQL data models and operational reporting pipelines (SQL Server, Python) to support analytical and operational use cases across business stakeholders.

Data Analyst, *MPundit Solutions* 11/2019 – 11/2021

- Built production ETL pipelines and dimensional models using SQL and Python to support self-service dashboards and feature health metrics.
- Conducted automated root-cause analysis on multi-structured data quality failures, standardizing data definitions across reporting tables.
- Collaborated with cross-functional stakeholders to deliver operational analytics and reliable data feeds under ambiguous product constraints.

- Worked independently with ambiguous requirements in a fast-paced environment, communicating technical data findings clearly to stakeholders and delivering pipeline solutions on time.

SKILLS

- **Languages & Querying: Python, SQL, KQL (familiarity), Java, PySpark, Spark SQL**
- **Data Platform & Lakehouse: Delta Lake, Databricks (Unity Catalog), Microsoft Fabric (concepts/OneLake), Snowflake, Redshift, BigQuery, ADLS Gen2, AWS S3**
- **AI & Agentic Systems: LangChain, LangGraph, AWS Bedrock, RAG Systems, Vector DBs (Pinecone, FAISS), Guardrails & LLM Evals, Prompt Orchestration**
- **Data Governance & Observability: Data Contracts, Semantic Modeling, dbt, Monte Carlo, CloudWatch, Data Lineage, SLA Alerting & Self-Healing Pipelines**
- **GenAI & ML:** LangChain, LangGraph, RAG, FAISS, Pinecone, Embeddings, Vector Databases, XGBoost, Scikit-learn, Pandas, NumPy
- **DevOps & Orchestration: Airflow, AWS Step Functions, EventBridge, Terraform, AWS CDK, Docker, Kubernetes, CI/CD (GitHub Actions), REST APIs**

PROJECTS

Autonomous Agentic Data Ingestion & RAG Platform

Designed an agentic knowledge retrieval and data transformation system using LangChain, LangGraph, and Pinecone to orchestrate schema discovery and natural language metric generation.

Technologies Python, LangGraph, LangChain, Pinecone, FAISS, Vector DB, REST APIs, Delta Lake

RAG Knowledge Assistant (Vector DB + Embeddings Pipeline)

Built a Generative AI and LLM-powered knowledge retrieval system using LangChain and Pinecone with metadata filtering and retrieval tuning, demonstrating hands-on GenAI, RAG, and ML integration experience.

Technologies: Python, PySpark, LangChain, LangGraph, FAISS, Pinecone, Embeddings, Vector DB

FIFA World Cup 2026 ML Prediction Engine

Designed an end-to-end sports analytics pipeline predicting match outcomes for 48 teams using a 4-model ensemble (XGBoost, Random Forest, Neural Network, Logistic Regression) with ELO ratings, FIFA rankings, and betting odds as features. Ran a Monte Carlo tournament simulator across 10,000 brackets to estimate win probabilities at every stage, integrating live World Cup 2026 match data via REST API to dynamically re-seed predictions in real time.

Technologies: Python, XGBoost, Scikit-learn, Pandas, NumPy, Monte Carlo Simulation, REST API, Git, HTML

EDUCATION

MS in Computer Science, University of Texas at Dallas — Dec 2023
BTech in Computer Science, SRM University — June 2021
DBA (Phd) in Business Intelligence & AI — June 2026